

Auteur: Thijs Van Schel
 Studierichting: Informatica
 Oefeningenreeks 5B nr. 2.2

$$\begin{aligned}
 & \int_{-\infty}^{+\infty} \frac{dx}{x^2 + 2x + 2}, \exists a \in]-\infty, +\infty[\\
 &= \int_{-\infty}^{+\infty} \frac{dx}{(x+1)^2 + 1} \\
 &= \int_{-\infty}^a \frac{dx}{(x+1)^2 + 1} + \int_a^{+\infty} \frac{dx}{(x+1)^2 + 1} \\
 &= \lim_{b \rightarrow -\infty} \int_b^a \frac{dx}{(x+1)^2 + 1} + \lim_{c \rightarrow +\infty} \int_a^c \frac{dx}{(x+1)^2 + 1} \\
 &= \lim_{b \rightarrow -\infty} [\text{Bgtan}(x+1)]_b^a + \lim_{c \rightarrow +\infty} [\text{Bgtan}(x+1)]_a^c \\
 &= \text{Bgtan}(a+1) + \frac{\pi}{2} - \text{Bgtan}(a+1) + \frac{\pi}{2} \\
 &= \frac{2\pi}{2} = \pi
 \end{aligned}$$

References